

ELEC2134 - Topic 2 Fourier Transform

Detailed Step-by-Step Tutorial Solutions - T3 2026

Fourier Transform convention used in the lecture:

$$X(\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t} dt, \quad x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega)e^{j\omega t} d\omega.$$

Slide 47 - Pass-Level Question 1(a)

Given

$$x(t) = e^{-3t}u(t-1).$$

Because $u(t-1) = 0$ for $t < 1$ and $u(t-1) = 1$ for $t \geq 1$,

$$X(\omega) = \int_1^{\infty} e^{-3t}e^{-j\omega t} dt = \int_1^{\infty} e^{-(3+j\omega)t} dt.$$

Integrating,

$$X(\omega) = \left[\frac{-e^{-(3+j\omega)t}}{3+j\omega} \right]_1^{\infty}.$$

Since $\Re\{3+j\omega\} = 3 > 0$, the exponential tends to zero as $t \rightarrow \infty$, hence

$$X(\omega) = \frac{e^{-(3+j\omega)}}{3+j\omega} = \frac{e^{-3}e^{-j\omega}}{3+j\omega}.$$

For the magnitude,

$$|X(\omega)| = \frac{e^{-3}|e^{-j\omega}|}{|3+j\omega|} = \frac{e^{-3}}{\sqrt{9+\omega^2}},$$

because $|e^{-j\omega}| = 1$.

For the phase,

$$\angle X(\omega) = \angle e^{-j\omega} - \angle(3+j\omega) = -\omega - \tan^{-1}\left(\frac{\omega}{3}\right).$$

$$\boxed{X(\omega) = \frac{e^{-3}e^{-j\omega}}{3+j\omega}}, \quad \boxed{|X(\omega)| = \frac{e^{-3}}{\sqrt{9+\omega^2}}},$$
$$\boxed{\angle X(\omega) = -\omega - \tan^{-1}\left(\frac{\omega}{3}\right)}.$$

Slide 47 - Pass-Level Question 1(b)

Given

$$x(t) = e^{-|t|}.$$

Split the integral at $t = 0$:

$$X(\omega) = \int_{-\infty}^0 e^te^{-j\omega t} dt + \int_0^{\infty} e^{-t}e^{-j\omega t} dt.$$

For the first integral,

$$\int_{-\infty}^0 e^{(1-j\omega)t} dt = \frac{1}{1-j\omega}.$$

For the second,

$$\int_0^{\infty} e^{-(1+j\omega)t} dt = \frac{1}{1+j\omega}.$$

Therefore,

$$X(\omega) = \frac{1}{1-j\omega} + \frac{1}{1+j\omega} = \frac{(1+j\omega) + (1-j\omega)}{1+\omega^2} = \frac{2}{1+\omega^2}.$$

Since this is real, positive, and even,

$$|X(\omega)| = \frac{2}{1+\omega^2}, \quad \angle X(\omega) = 0.$$

$$\boxed{X(\omega) = \frac{2}{1+\omega^2}}, \quad \boxed{|X(\omega)| = \frac{2}{1+\omega^2}}, \quad \boxed{\angle X(\omega) = 0}.$$

Slide 47 - Pass-Level Question 2(a)

Given

$$X(\omega) = e^{-2\omega} u(\omega).$$

Use the inverse Fourier Transform:

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{j\omega t} d\omega.$$

Since $u(\omega) = 0$ for $\omega < 0$,

$$x(t) = \frac{1}{2\pi} \int_0^{\infty} e^{-2\omega} e^{j\omega t} d\omega = \frac{1}{2\pi} \int_0^{\infty} e^{-(2-jt)\omega} d\omega.$$

Thus,

$$x(t) = \frac{1}{2\pi} \left[\frac{-e^{-(2-jt)\omega}}{2-jt} \right]_0^{\infty} = \frac{1}{2\pi(2-jt)}.$$

Multiplying numerator and denominator by $2+jt$,

$$x(t) = \frac{2+jt}{2\pi(4+t^2)}.$$

$$\boxed{x(t) = \frac{1}{2\pi(2-jt)} = \frac{2+jt}{2\pi(4+t^2)}}.$$

Slide 47 - Pass-Level Question 2(b)

Given

$$X(\omega) = e^{-|\omega|}.$$

Since $X(\omega)$ is real and even,

$$x(t) = \frac{1}{\pi} \int_0^{\infty} e^{-\omega} \cos(\omega t) d\omega.$$

Use the standard integral

$$\int_0^{\infty} e^{-a\omega} \cos(b\omega) d\omega = \frac{a}{a^2 + b^2}, \quad a > 0.$$

With $a = 1$ and $b = t$,

$$\boxed{x(t) = \frac{1}{\pi(1+t^2)}}.$$

Slide 47 - Pass-Level Question 3: Inverse Transform from Magnitude and Phase

The magnitude is unity over $-2 < \omega < 2$, and the phase is

$$\angle X(\omega) = \begin{cases} +\frac{\pi}{2}, & -2 < \omega < 0, \\ -\frac{\pi}{2}, & 0 < \omega < 2. \end{cases}$$

Therefore,

$$X(\omega) = \begin{cases} j, & -2 < \omega < 0, \\ -j, & 0 < \omega < 2, \\ 0, & \text{otherwise.} \end{cases}$$

Apply the inverse Fourier Transform:

$$x(t) = \frac{1}{2\pi} \left[\int_{-2}^0 j e^{j\omega t} d\omega - \int_0^2 j e^{j\omega t} d\omega \right].$$

Since

$$\int j e^{j\omega t} d\omega = \frac{e^{j\omega t}}{t},$$

we obtain

$$x(t) = \frac{1}{2\pi t} \left[(1 - e^{-j2t}) - (e^{j2t} - 1) \right].$$

Thus,

$$x(t) = \frac{1}{2\pi t} \left[2 - e^{j2t} - e^{-j2t} \right].$$

Using

$$e^{j2t} + e^{-j2t} = 2 \cos(2t),$$

we obtain

$$\boxed{x(t) = \frac{1 - \cos(2t)}{\pi t}, \quad t \neq 0.}$$

At $t = 0$,

$$x(0) = \lim_{t \rightarrow 0} \frac{1 - \cos(2t)}{\pi t} = 0.$$

$$\boxed{x(t) = \frac{1 - \cos(2t)}{\pi t}, \quad t \neq 0,} \quad \boxed{x(0) = 0}.$$

Slide 48 - Pass-Level Question 4(a)

Given

$$x(t) = A \sin(\omega_0 t) + A \cos(2\omega_0 t).$$

Use the standard transform pairs

$$\cos(\omega_0 t) \xleftrightarrow{\mathcal{F}} \pi [\delta(\omega - \omega_0) + \delta(\omega + \omega_0)],$$

$$\sin(\omega_0 t) \xleftrightarrow{\mathcal{F}} -j\pi\delta(\omega - \omega_0) + j\pi\delta(\omega + \omega_0).$$

Hence,

$$\boxed{X(\omega) = -jA\pi\delta(\omega - \omega_0) + jA\pi\delta(\omega + \omega_0) + A\pi\delta(\omega - 2\omega_0) + A\pi\delta(\omega + 2\omega_0)}.$$

The magnitude spectrum has four lines at

$$\omega = \pm\omega_0, \quad \omega = \pm 2\omega_0,$$

each of magnitude $A\pi$.

Slide 48 - Pass-Level Question 4(b)

Given

$$x(t) = \sin(2\pi t) + \cos(3\pi t).$$

For the sine term, the spectral lines are at $\omega = \pm 2\pi$ with magnitude π and phases

$$\angle X(+2\pi) = -\frac{\pi}{2}, \quad \angle X(-2\pi) = +\frac{\pi}{2}.$$

For the cosine term, spectral lines are at $\omega = \pm 3\pi$, each with magnitude π and phase 0.

$$\text{Spectral locations: } \omega = \pm 2\pi, \pm 3\pi.$$

Sine lines: magnitude π , phases $\mp\pi/2$ at positive/negative frequency.

Cosine lines: magnitude π , phase 0 at both frequencies.

Slide 48 - Pass-Level Question 4(c)

Given

$$x(t) = 5 \cos(3\pi t) \cos(\pi t).$$

Use

$$\cos A \cos B = \frac{1}{2} [\cos(A + B) + \cos(A - B)].$$

Therefore,

$$x(t) = \frac{5}{2} [\cos(4\pi t) + \cos(2\pi t)].$$

Using the cosine transform pair,

$$X(\omega) = \frac{5\pi}{2} [\delta(\omega - 4\pi) + \delta(\omega + 4\pi) + \delta(\omega - 2\pi) + \delta(\omega + 2\pi)].$$

Hence the four spectral lines are at

$$\omega = \pm 2\pi, \pm 4\pi$$

with magnitude

$$\frac{5\pi}{2}.$$

Slide 48 - Pass-Level Question 5(a)

Given

$$F(\omega) = \delta(\omega - 10) - 2\delta(\omega) - 3\delta(\omega + 30).$$

The magnitude spectrum has line magnitudes

$$|F(10)| = 1, \quad |F(0)| = 2, \quad |F(-30)| = 3.$$

Use the inverse Fourier Transform:

$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) e^{j\omega t} d\omega.$$

Applying the sifting property of the delta function,

$$f(t) = \frac{1}{2\pi} [e^{j10t} - 2 - 3e^{-j30t}].$$

The result is complex-valued because the specified spectrum does not satisfy conjugate symmetry.

Slide 48 - Pass-Level Question 5(b)

Given

$$f(t) = \frac{1}{2} + \frac{2}{\pi} \sin t + \frac{2}{3\pi} \sin 3t + \frac{2}{5\pi} \sin 5t.$$

The DC term gives

$$\frac{1}{2} \xleftrightarrow{\mathcal{F}} \pi \delta(\omega).$$

Using the sine transform pair for each sinusoid,

$$\begin{aligned} F(\omega) = & \pi \delta(\omega) - 2j \delta(\omega - 1) + 2j \delta(\omega + 1) \\ & - \frac{2j}{3} \delta(\omega - 3) + \frac{2j}{3} \delta(\omega + 3) \\ & - \frac{2j}{5} \delta(\omega - 5) + \frac{2j}{5} \delta(\omega + 5). \end{aligned}$$

Magnitude lines:

$$\pi \text{ at } 0; \quad 2 \text{ at } \pm 1; \quad \frac{2}{3} \text{ at } \pm 3; \quad \frac{2}{5} \text{ at } \pm 5.$$

Positive-frequency sine lines have phase $-\pi/2$; negative-frequency sine lines have phase $+\pi/2$.

Slide 48 - Pass-Level Question 6: RC Circuit

Given

$$R = 2 \, \Omega, \quad C = 1 \, \text{F}, \quad v_i(t) = 2e^{-3t}u(t),$$

with zero initial conditions.

The input transform is

$$V_i(\omega) = \frac{2}{3 + j\omega}.$$

For the capacitor,

$$Z_C = \frac{1}{j\omega C} = \frac{1}{j\omega}.$$

Using voltage division with output across the capacitor,

$$H(j\omega) = \frac{Z_C}{R + Z_C} = \frac{1/(j\omega)}{2 + 1/(j\omega)} = \frac{1}{1 + j2\omega}.$$

Therefore,

$$V_o(\omega) = H(j\omega)V_i(\omega) = \frac{2}{(3 + j\omega)(1 + j2\omega)}.$$

Since

$$1 + j2\omega = 2 \left(\frac{1}{2} + j\omega \right),$$

we obtain

$$V_o(\omega) = \frac{1}{(j\omega + 3)(j\omega + 0.5)}.$$

Partial fractions give

$$V_o(\omega) = 0.4 \left[\frac{1}{j\omega + 0.5} - \frac{1}{j\omega + 3} \right].$$

Using

$$\frac{1}{a + j\omega} \xleftrightarrow{\mathcal{F}^{-1}} e^{-at}u(t),$$

we obtain

$$v_o(t) = 0.4 \left(e^{-0.5t} - e^{-3t} \right) u(t).$$

Slide 49 - Higher-Level Question 1

Use the base pair

$$x(t) = \begin{cases} 1, & |t| \leq 1, \\ 0, & \text{otherwise,} \end{cases} \quad X(\omega) = \frac{2 \sin \omega}{\omega}.$$

(a) Shifted rectangle

If

$$y(t) = x(t - 1),$$

then by the time-shift property,

$$Y(\omega) = e^{-j\omega} X(\omega).$$

Hence,

$$Y(\omega) = e^{-j\omega} \frac{2 \sin \omega}{\omega}.$$

(b) Windowed cosine

If

$$y(t) = x(t) \cos(\pi t),$$

then the modulation property gives

$$Y(\omega) = \frac{1}{2} [X(\omega - \pi) + X(\omega + \pi)].$$

Substituting the known $X(\omega)$,

$$Y(\omega) = \frac{\sin(\omega - \pi)}{\omega - \pi} + \frac{\sin(\omega + \pi)}{\omega + \pi}.$$

(c) Truncated ramp

If

$$y(t) = t x(t),$$

then

$$tx(t) \xleftrightarrow{\mathcal{F}} j \frac{dX(\omega)}{d\omega}.$$

Therefore,

$$Y(\omega) = j \frac{d}{d\omega} \left(\frac{2 \sin \omega}{\omega} \right).$$

Using the quotient rule,

$$\frac{d}{d\omega} \left(\frac{2 \sin \omega}{\omega} \right) = 2 \frac{\omega \cos \omega - \sin \omega}{\omega^2}.$$

Thus,

$$Y(\omega) = \frac{2j [\omega \cos \omega - \sin \omega]}{\omega^2} = \frac{2j}{\omega} \left(\cos \omega - \frac{\sin \omega}{\omega} \right).$$

Slide 49 - Higher-Level Question 2: Gaussian

Given

$$x(t) = e^{-at^2}, \quad a > 0.$$

The transform is

$$X(\omega) = \int_{-\infty}^{\infty} e^{-at^2} e^{-j\omega t} dt.$$

Differentiate with respect to ω :

$$\frac{dX}{d\omega} = \int_{-\infty}^{\infty} (-jt) e^{-at^2} e^{-j\omega t} dt.$$

Since

$$\frac{d}{dt} e^{-at^2} = -2at e^{-at^2},$$

we have

$$te^{-at^2} = -\frac{1}{2a} \frac{d}{dt} e^{-at^2}.$$

Therefore,

$$\frac{dX}{d\omega} = \frac{j}{2a} \int_{-\infty}^{\infty} \frac{d}{dt} (e^{-at^2}) e^{-j\omega t} dt.$$

Integrating by parts, the boundary term vanishes because $e^{-at^2} \rightarrow 0$ as $|t| \rightarrow \infty$. Hence,

$$\frac{dX}{d\omega} = -\frac{\omega}{2a} X(\omega).$$

Thus,

$$\frac{dX}{X} = -\frac{\omega}{2a} d\omega.$$

Integrating,

$$\ln X = -\frac{\omega^2}{4a} + C,$$

so

$$X(\omega) = K e^{-\omega^2/(4a)}.$$

At $\omega = 0$,

$$X(0) = \int_{-\infty}^{\infty} e^{-at^2} dt = \sqrt{\frac{\pi}{a}}.$$

Therefore,

$$X(\omega) = \sqrt{\frac{\pi}{a}} e^{-\omega^2/(4a)}.$$

Effect of increasing a

The time-domain signal is

$$x(t) = e^{-at^2}.$$

Its characteristic width is proportional to $1/\sqrt{a}$. Therefore increasing a makes the time-domain waveform narrower.

In frequency,

$$X(\omega) = \sqrt{\frac{\pi}{a}} e^{-\omega^2/(4a)},$$

whose characteristic width is proportional to $\sqrt{a}$. Therefore increasing a makes the spectrum wider.

$$a \uparrow \Rightarrow \text{narrower in time, wider in frequency.}$$

Problem-solving reminder: identify structure and the appropriate Fourier property before starting an integral.